Subject: Mathematics | **WEEKLY CHAPTER TEST – 05 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

In a single throw of two dice, the probability of getting 6 as a product of the two numbers obtained is:

(a) $\frac{1}{9}$ (b) $\frac{2}{9}$

(c) $\frac{1}{6}$ (d) $\frac{5}{36}$

Working & Calculation:

- When two dice are thrown together, total elementary outcomes: $n(S) = 6 \times 6 = 36$.
- Let E be the event that the product of the two numbers equals 6:

$$E = \{(1,6),(2,3),(3,2),(6,1)\} \implies n(E) = 4$$

- Therefore:

$$P(\text{product is } 6) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{9}$

Question 2 [1 Mark]

The king, queen, and jack of clubs are removed from a deck of 52 cards. A card is drawn from the remaining cards. The probability of getting a club card is:

(a) $\frac{10}{49}$ (b) $\frac{13}{49}$

(c) $\frac{1}{4}$ (d) $\frac{3}{49}$

Working & Calculation:

- Number of cards removed: 3 cards (King, Queen, Jack of clubs).
- Total remaining cards: $n(S') = 52 - 3 = 49$.
- A standard deck contains 13 clubs. With 3 club cards removed, remaining clubs $= 13 - 3 = 10$.
- Therefore:

$$P(\text{a club card}) = \frac{n(\text{clubs remaining})}{n(S')} = \frac{10}{49}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{10}{49}$

Question 3 [1 Mark]

A box contains 5 apples, 6 oranges, and x bananas. If the probability of selecting an apple from the box is $\frac{1}{3}$, then the number of bananas in the box is:

(a) 3 **(b) 4**

(c) 5 **(d) 6**

Working & Calculation:

- Total fruits: $n(S) = 5 + 6 + x = 11 + x$.
- Given $P(\text{apple}) = \frac{1}{3} \implies \frac{5}{11+x} = \frac{1}{3} \implies 11 + x = 15 \implies x = 4$.

CORRECT ANSWER: OPTION (b) $\longrightarrow$ 4

Question 4 [1 Mark]

A letter is chosen at random from the letters of the word 'ASSASSINATION'. The probability that the letter chosen is a vowel is:

(a) $\frac{6}{13}$ **(b) $\frac{7}{13}$**

(c) $\frac{5}{13}$ **(d) $\frac{4}{13}$**

Working & Calculation:

- Total letters in 'ASSASSINATION': $n(S) = 13$.
- Vowels are A, A, I, A, I, O $\implies n(\text{vowels}) = 6 \implies P(\text{vowel}) = \frac{6}{13}$.

CORRECT ANSWER: OPTION (a) $\longrightarrow$ $\frac{6}{13}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): When two coins are tossed simultaneously, the probability of getting at least one head is $\frac{3}{4}$.

Reason (R): The sample space of tossing two coins is $\{HH, HT, TH, TT\}$, containing 4 equally likely outcomes.

Options:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Reason (R):** Two fair coins yield S = {HH, HT, TH, TT} $\implies n(S) = 4$ outcomes. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** Favourable outcomes with at least one head: {HT, TH, HH} $\implies n(E) = 3 \implies P = \frac{3}{4}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** (R) provides the complete sample space used to compute (A). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

A die is thrown once. Find the probability of getting: (i) a number divisible by 3 (ii) a composite number.

Sample Space: When a standard die is thrown once, $S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$.

(i) A number divisible by 3 [1 Mark]

Numbers divisible by 3 on a die are $E_1 = \{3, 6\} \implies n(E_1) = 2$:

$$P(\text{divisible by } 3) = \frac{n(E_1)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

Answer (i): $\mathbf{P(\text{divisible by } 3) = \frac{1}{3}}$

(ii) A composite number [1 Mark]

A composite number has more than two distinct positive factors. Among $\{1, 2, 3, 4, 5, 6\}$:

- 1 is neither prime nor composite; $2, 3, 5$ are prime numbers.
- Composite numbers are $E_2 = \{4, 6\} \implies n(E_2) = 2$.

$$P(\text{a composite number}) = \frac{n(E_2)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

Answer (ii): $\mathbf{P(\text{composite number}) = \frac{1}{3}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result (i.e., three heads or three tails), and loses otherwise.
(i) Write all possible outcomes of the game.
(ii) Calculate the probability that Hanif will lose the game.
(iii) What is the probability that he gets at least one head?

(i) All possible outcomes of the game [1 Mark]

The complete sample space S consists of $2^3 = 8$ outcomes:

$$S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\} \implies n(S) = 8$$

Answer (i): $\mathbf{S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad n(S) = 8}$

(ii) Probability that Hanif will lose the game [1 Mark]

Hanif wins if outcomes are HHH or TTT (2 winning outcomes). Thus, losing outcomes are $8 - 2 = 6$:

$$E_{\text{lose}} = \{\text{HHT, HTH, HTT, THH, THT, TTH}\} \implies P(\text{lose}) = \frac{6}{8} = \frac{3}{4}$$

Answer (ii): $\mathbf{P(\text{Hanif loses}) = \frac{3}{4}}$

(iii) Probability that he gets at least one head [1 Mark]

Outcomes with at least one head include all outcomes except TTT ($8 - 1 = 7$ outcomes):

$$P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - \frac{1}{8} = \frac{7}{8}$$

Answer (iii): $\mathbf{P(\text{at least one head}) = \frac{7}{8}}$

Question 8 [3 Marks]

A bag contains 6 red, 5 black, and 4 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is:
(i) white (ii) red or black (iii) neither red nor black.

Sample Space: Total balls in the bag: $n(S) = 6 + 5 + 4 = 15$.

(i) A white ball [1 Mark]

Number of white balls $= 4 \implies P(\text{white}) = \dfrac{n(\text{white})}{n(S)} = \dfrac{4}{15}$.

Answer (i): $\mathbf{P(\text{white}) = \dfrac{4}{15}}$

(ii) A red or black ball [1 Mark]

Number of red or black balls: $6 + 5 = 11 \implies P(\text{red or black}) = \dfrac{11}{15}$.

Answer (ii): $\mathbf{P(\text{red or black}) = \dfrac{11}{15}}$

(iii) Neither red nor black [1 Mark]

A ball that is neither red nor black must be white (4 balls): $P(\text{neither red nor black}) = 1 - \dfrac{11}{15} = \dfrac{4}{15}$.

Answer (iii): $\mathbf{P(\text{neither red nor black}) = \dfrac{4}{15}}$

Question 9 [3 Marks]

Two customers, Shyam and Ekta, are visiting a particular shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on:
(i) the same day? (ii) consecutive days? (iii) different days?

Sample Space: Days available are {Tue, Wed, Thu, Fri, Sat} (5 days). Total pairs: $n(S) = 5 \times 5 = 25$.

(i) The same day [1 Mark]

Favourable pairs are $E_1 = \{(\text{Tue, Tue}), (\text{Wed, Wed}), (\text{Thu, Thu}), (\text{Fri, Fri}), (\text{Sat, Sat})\} \implies n(E_1) = 5$:

$$P(\text{same day}) = \frac{5}{25} = \frac{1}{5}$$

Answer (i): $\mathbf{P(\text{same day}) = \dfrac{1}{5}}$

(ii) Consecutive days [1 Mark]

Consecutive pairs include 4 pairs where Shyam precedes Ekta and 4 pairs where Ekta precedes Shyam:

- Shyam precedes Ekta: $\{(\text{Tue, Wed}), (\text{Wed, Thu}), (\text{Thu, Fri}), (\text{Fri, Sat})\}$ (4 pairs).
- Ekta precedes Shyam: $\{(\text{Wed, Tue}), (\text{Thu, Wed}), (\text{Fri, Thu}), (\text{Sat, Fri})\}$ (4 pairs).

Total favourable outcomes $n(E_2) = 4 + 4 = 8 \implies P(\text{consecutive days}) = \dfrac{8}{25}$.

Answer (ii): $\mathbf{P(\text{consecutive days}) = \dfrac{8}{25}}$

(iii) Different days [1 Mark]

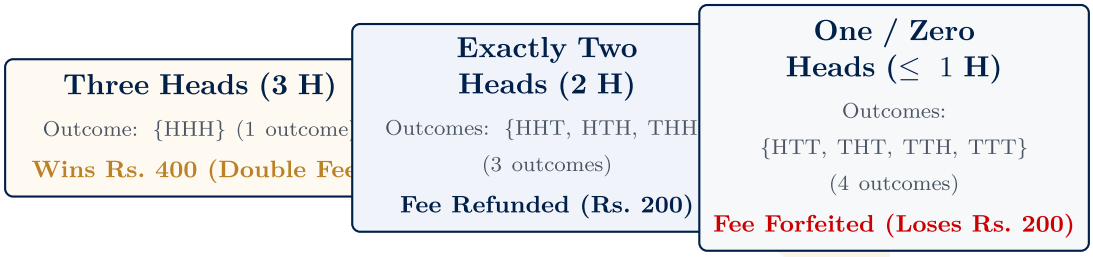
Visiting on different days is the complement of visiting on the same day: $P(\text{different days}) = 1 - \dfrac{1}{5} = \dfrac{4}{5}$.

Answer (iii): $\mathbf{P(\text{different days}) = \dfrac{4}{5}}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10 (Case Study Analysis) [4 Marks]

A school fair organizer sets up a game called "Lucky Trio" involving the simultaneous toss of three fair coins. The entry fee is Rs. 200. The payout rules are structured as follows:
· If a player gets three heads, they win double their entry fee (Rs. 400).
· If a player gets exactly two heads, their entry fee of Rs. 200 is refunded.
· If a player gets one head or zero heads, their entry fee is forfeited (they lose).
Shikha enters the game by paying Rs. 200. Based on the above scenario, answer the following:



Probability Model: Total outcomes for tossing 3 fair coins: $n(S) = 2^3 = 8$.

(i) Write down all 8 outcomes in the sample space [1 Mark]

The complete sample space showing all sequences of Heads (H) and Tails (T):

$$S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\}$$

Answer (i): $\mathbf{S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad n(S) = 8}$

(ii) Probability that Shikha wins double the entry fee [1 Mark]

Winning double requires obtaining 3 heads $\implies E_1 = \{\text{HHH}\} \implies n(E_1) = 1$:

$$P(\text{double fee}) = \frac{n(E_1)}{n(S)} = \frac{1}{8}$$

Answer (ii): $\mathbf{P(\text{wins double entry fee}) = \frac{1}{8}}$

(iii) Probability that Shikha gets her entry fee refunded [1 Mark]

Refund condition is getting exactly 2 heads $\implies E_2 = \{\text{HHT, HTH, THH}\} \implies n(E_2) = 3$:

$$P(\text{refund}) = \frac{n(E_2)}{n(S)} = \frac{3}{8}$$

Answer (iii): $\mathbf{P(\text{entry fee refunded}) = \frac{3}{8}}$

(iv) Probability that Shikha loses her entry fee [1 Mark]

Loss condition is getting one head or zero heads $\implies E_3 = \{\text{HTT, THT, TTH, TTT}\} \implies n(E_3) = 4$:

$$P(\text{loses entry fee}) = \frac{n(E_3)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Answer (iv): $\mathbf{P(\text{loses entry fee}) = \frac{1}{2}}$

— END OF WORKED SOLUTIONS (TEST–05) · MATHS MANTHAN TUITION CLASSES —